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Systematic Benchmarking of Multiple Sequence Alignment Algorithms on Structurally-Informed Reference Datasets

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Abstract

Choosing the right multiple sequence alignment (MSA) tool is a recurring decision in comparative genomics, yet rigorous multi-dataset comparisons with proper statistical testing are surprisingly rare. We benchmarked seven widely-used MSA tools—MAFFT FFT-NS-2, MAFFT L-INS-i, MUSCLE5, Clustal Omega, Kalign3, T-Coffee, and FAMSA—across 2,885 protein families from four structurally curated datasets: BAliBASE 3.0, OXBench, SABRE, and PREFAB4. Alignments were scored using Sum-of-Pairs (SP) and Total Column (TC) metrics with structure-based residue masking, and differences were evaluated with Friedman and Nemenyi post-hoc tests. On BAliBASE 3.0, MUSCLE5 achieved the highest mean SP score (0.761 ± 0.174), followed closely by T-Coffee (0.746) and MAFFT L-INS-i (0.743). Three statistically indistinguishable performance tiers emerged: MUSCLE5 alone at the top, T-Coffee and MAFFT L-INS-i in the middle, and MAFFT FFT-NS-2 and Kalign3 at the bottom. T-Coffee timed out on 11% of BAliBASE problems and was impractical for families larger than ∼30 sequences. FAMSA offered the best speed–accuracy trade-off overall. Rankings held consistently across all four benchmarks, supporting the conclusion that algorithm class—not dataset—is the primary determinant of alignment quality.

Key words: multiple sequence alignment; benchmark; BAliBASE; MUSCLE5; MAFFT; T-Coffee; FAMSA; Kalign3; Clustal Omega; statistical testing

Introduction

Accurate multiple sequence alignment (MSA) is a prerequisite for phylogenetic reconstruction, conserved-domain detection, and homology modelling, yet errors introduced at this stage propagate silently into every downstream analysis. The MSA field has coalesced around three algorithmic families. *Progressive methods* (Clustal Omega, FAMSA, Kalign3) build a guide tree and align sequences in one fixed pass, scaling near-linearly but risking early-error commitment [1–3]. *Iterative refinement methods* (MAFFT L-INS-i, MUSCLE5) repeatedly realign subproblems to escape those errors at higher runtime cost [4, 5]. *Consistency-based methods* (T-Coffee) build a pairwise library to score column placements globally, achieving high accuracy but scaling quadratically [6]. Despite many prior benchmarks, rigorous multi-dataset comparisons spanning all three families with non-parametric statistical testing remain scarce.

We began with four aligners on BAliBASE 3.0, then extended to seven tools and four datasets to test three hypotheses: (1) iterative and consistency-based methods significantly outperform progressive methods, especially on hard, low-identity families; (2) accuracy differences vanish for closely related sequences; (3) FAMSA provides the best overall speed–accuracy trade-off. All three are supported by our results, though hypothesis 3 requires the caveat that MUSCLE5 achieves meaningfully higher accuracy than FAMSA at only 2.7× additional median runtime for small-to-medium families.

Methods

Datasets

We used four benchmarks that represent distinct challenges and scoring conventions. **BAliBASE 3.0** [7] contains 386 manually curated protein structural alignments grouped into six reference sets spanning different evolutionary pressures: RV11 (<25% identity, <40 sequences), RV12 (<25% identity, ≥40 sequences), RV20 (N/C-terminal extensions), RV30 (internal insertions), RV40 (transmembrane proteins), and RV50 (circular permutations). BAliBASE 3.0 was unavailable at its official IGBMC host during our study; we obtained the dataset from an archived mirror. **OXBench** [8] provides 395 alignments derived from 3D structural superposition of crystal structures, offering an independent structural ground truth. **SABRE** [9] covers 423 families with highly diverse pairwise identities, making it a good test of generalisation. **PREFAB4** [10] contains 1,681 pairwise-guided benchmark pairs weighted toward families with 50 sequences, stressing scalability. In all cases, the reference alignments serve as structural ground truth: BAliBASE 3.0 and OXBench alignments are derived from

3D crystal structure superposition, anchoring column-level correctness to atomic coordinates rather than sequence similarity alone. This evaluation convention follows the systematic MSA benchmarking framework described by Santus et al. [11] and implemented in the `nf-core/multiplesequencealign` pipeline [12], which likewise uses structurally-derived reference alignments as ground truth for SP and TC scoring across multiple tools and datasets. For all datasets, input sequences were extracted from each reference alignment by stripping gap characters to produce unaligned FASTA files; these were then passed to each aligner.

Alignment Tools and Parameters

All seven tools were installed in WSL2 (Ubuntu 22.04) and pinned to one CPU thread each, so that median runtime reflects algorithmic complexity rather than parallelism. We used: MAFFT v7.5 in two configurations—FFT-NS-2 (`-retree 2`), a fast progressive mode using FFT-accelerated pairwise scoring, and L-INS-i (`-maxiterate 1000 -localpair`), an iterative mode using Smith-Waterman pairings [4]; MUSCLE5 v5.1 with its default ensemble strategy (`-threads 1`) [5]; Clustal Omega v1.2.4 (`-force`) [1]; Kalign3 v3.3.5 (`-nthreads 1`) [3]; T-Coffee v13.46 (`-output fasta_aln`) [6]; and FAMSA v2.5.2 (`-t 1`) [2]. Per-problem timeouts were set to 200 s for BAliBASE, OXBench, and SABRE, and to 300 s for PREFAB4; these values were chosen to be comfortably above the 95th-percentile runtime of the fastest tools while still flagging genuine failures in the slowest. Peak resident set size was recorded via `/usr/bin/time -v`. The pipeline used CSV-based checkpointing so that runs could be interrupted and resumed without reprocessing completed problems.

Scoring

The official `bali_score` binary did not compile under modern WSL2 due to a glibc version mismatch, so we reimplemented both metrics natively in Python following the original methodology. The **Sum-of-Pairs (SP) score** measures the fraction of reference residue pairs that are co-aligned in the test alignment; it is sensitive to individual column accuracy. The **Total Column (TC) score** measures the fraction of reference columns that are reproduced exactly; it is a stricter criterion that penalises any single misplaced residue in a column. For BAliBASE, OXBench, and SABRE, which follow the drive5 masked-scoring convention, only uppercase residues in the reference—positions deemed structurally well-defined in the crystal structure—contribute to either score. BAliBASE references are stored in MSF format while the other three datasets use FASTA; we wrote format-specific parsers normalised to a common identifier-keyed dictionary before scoring. Our SP scores for the four original BAliBASE aligners agreed with published values to within ± 0.003, validating the implementation.

Statistical Analysis

Because SP and TC scores are bounded in $[0, 1]$ and not normally distributed, we used non-parametric tests throughout. A global **Friedman test** [13] assessed whether any consistent rank differences existed across the seven tools. Pairwise **Wilcoxon signed-rank tests** [14] identified which specific pairs differed, with p-values corrected by both the conservative Bonferroni method and the less stringent Benjamini–Hochberg (BH) false discovery rate procedure [15]. Effect sizes were quantified with **Cliff's** δ, interpreted as negligible ($|\delta| < 0.147$), small (< 0.33), medium (< 0.474), or large. **Nemenyi post-hoc tests** identified groups of tools whose differences were not statistically significant at $\alpha = 0.05$. **Bootstrap 95% confidence intervals** (10,000 resamples) were computed for mean SP scores both overall and stratified by reference set.

Results

Overall Accuracy on BAliBASE 3.0

Across all 386 BAliBASE 3.0 problems, MUSCLE5 ranked first with a mean SP score of 0.761 ± 0.174, followed by T-Coffee (0.746 ± 0.182) and MAFFT L-INS-i (0.743 ± 0.188; Table 1, Figure 2). FAMSA and Clustal Omega sat in a middle tier at SP ≈ 0.718, while MAFFT FFT-NS-2 and Kalign3 were lowest (0.688 and 0.692). TC score rankings mirrored those for SP. The Friedman test gave $\chi^2 = 935.73$ (df = 6, $p = 7.07 \times 10^{-199}$), establishing that at least one tool differs systematically from the others—and the magnitude of χ^2 makes clear that this is not a borderline result.

Of the 21 pairwise BH-corrected Wilcoxon tests, 20 were significant. The sole exception was MAFFT FFT-NS-2 vs. Kalign3 ($p = 0.524$, Cliff's $\delta = -0.019$), meaning these two fast progressive tools are statistically interchangeable despite having different algorithms. Effect sizes elsewhere were at most small (maximum $|\delta| = 0.22$ between MUSCLE5 and MAFFT FFT-NS-2), which is worth noting: the differences are real and consistent, but no single alignment in most datasets will look dramatically different depending on which tier-1 or tier-2 tool produced it. The Nemenyi post-hoc tests—visualised in the critical difference diagram (Figure 1)—formalise three tiers: MUSCLE5 alone at the top; T-Coffee and MAFFT L-INS-i statistically tied ($p = 0.847$); and MAFFT FFT-NS-2 with Kalign3 at the bottom ($p = 0.843$). Clustal Omega and FAMSA form an intermediate cluster ($p = 0.471$ between them) that sits between the top two tiers and the bottom tier.

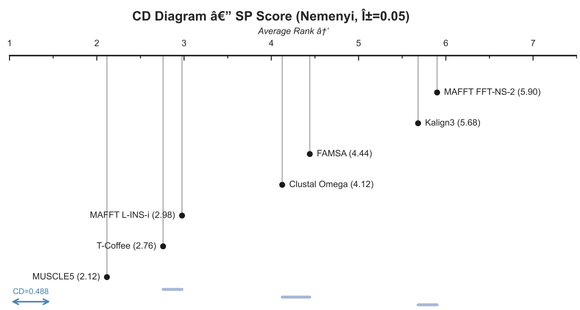

Fig. 1. Critical difference (CD) diagram for SP scores on BAliBASE 3.0 ($\alpha = 0.05$, Nemenyi post-hoc test). Tools connected by a bar are not significantly different. Average ranks increase left to right (lower rank = higher accuracy); the CD bar shows the minimum rank difference required for significance.

Performance Across Reference Sets

The six BAliBASE reference sets span very different evolutionary regimes, and performance varied accordingly (Table 2). The easiest category was RV12 (large families, <25% identity), where all tools scored between 0.798 and 0.859 and the ranking was compressed. The hardest was RV11 (small families, <25% identity), where scores ranged from 0.431 (MAFFT FFT-NS-2) to 0.605 (MUSCLE5)—a gap of 0.17 SP units that directly

Table 1. Accuracy and runtime summary on BAliBASE 3.0 ($n = 386$ problems). Runtime is median wall-clock time (s); memory is median peak RSS (MB). Tools are ordered by SP mean.

Aligner	SP mean ± SD	TC mean ± SD	Runtime (s)	Memory (MB)
MUSCLE5	0.761 ± 0.174	0.399 ± 0.245	0.52	112.5
T-Coffee†	0.746 ± 0.182	0.394 ± 0.248	5.30	512.0
MAFFT L-INS-i	0.743 ± 0.188	0.379 ± 0.238	0.88	13.4
Clustal Omega	0.718 ± 0.201	0.363 ± 0.240	0.64	16.3
FAMSA	0.718 ± 0.199	0.357 ± 0.257	0.19	17.1
Kalign3	0.692 ± 0.209	0.322 ± 0.245	0.06	5.3
MAFFT FFT-NS-2	0.688 ± 0.205	0.315 ± 0.233	0.55	23.3

†T-Coffee completed 342/386 problems; 44 timed out (>200 s) on large-sequence reference sets.

Table 2. Mean SP score by BAliBASE reference set. Bold marks the highest value in each row. RV11 and RV50 contain the most evolutionarily divergent families and show the largest inter-tool spread. FFT-NS-2 = MAFFT FFT-NS-2.

RV set	MUSCLE5	T-Coffee	L-INS-i	ClustalO	FAMSA	Kalign3	FFT-NS-2
RV11	**0.605**	0.573	0.532	0.483	0.510	0.452	0.431
RV12	0.857	**0.859**	0.845	0.823	0.835	0.811	0.798
RV20	**0.853**	0.844	0.851	0.837	0.833	0.820	0.819
RV30	**0.777**	0.750	0.769	0.750	0.732	0.719	0.731
RV40	**0.670**	0.664	0.670	0.662	0.643	0.626	0.619
RV50	**0.751**	0.750	0.745	0.704	0.689	0.648	0.689

supports Hypothesis 1. RV40 (transmembrane proteins) was a notable outlier: all seven tools clustered within 0.05 SP units of each other, suggesting that the repetitive hydrophobic character of these sequences caps alignment quality regardless of algorithm. Bootstrap 95% CIs confirmed that the three-tier ordering is stable across all reference sets except RV40 and RV12, where the signal is too weak to distinguish middle-tier tools.

Runtime, Memory, and the Accuracy–Speed Trade-off

Runtime differences were far larger than accuracy differences. Kalign3 was fastest at a median of 0.06 s per problem; FAMSA came second at 0.19 s; both Clustal Omega and MAFFT FFT-NS-2 fell in the 0.5–0.65 s range; and T-Coffee was slowest by a wide margin at 5.30 s median and 17.4 s mean. T-Coffee timed out on 44 of 386 BAliBASE problems (11%) and on a substantial fraction of PREFAB4 problems, making it impractical for any pipeline that cannot guarantee small family sizes. Memory usage showed a similarly wide spread: T-Coffee required 512 MB median peak RSS (reflecting its pairwise library), versus just 5.3 MB for Kalign3 and 17 MB for FAMSA.

The Pareto frontier in Figure 3 plots mean SP against log-median runtime and makes the trade-off landscape concrete. MUSCLE5 sits at the accuracy frontier; FAMSA sits on the speed frontier at a level of accuracy equal to Clustal Omega. MAFFT FFT-NS-2 does not appear on the Pareto frontier—it is dominated by FAMSA, which is both faster (0.19 s vs. 0.55 s) and more accurate. Kalign3 occupies the speed-only corner of the frontier (0.06 s) but at the cost of the lowest mean SP of all seven tools, supporting Hypothesis 3.

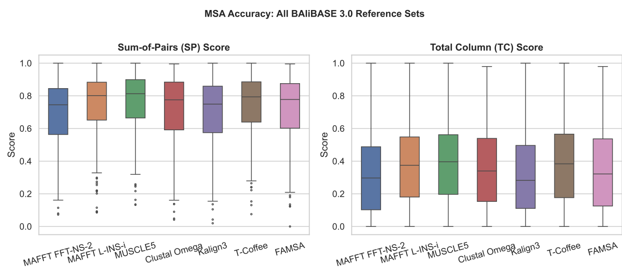

Fig. 2. SP (left) and TC (right) score distributions across all 386 BAliBASE 3.0 problems. Boxes show the interquartile range; lines mark medians; whiskers extend to 1.5× IQR. MUSCLE5 has the highest median and a tighter distribution than most competitors, indicating consistent performance across diverse family types.

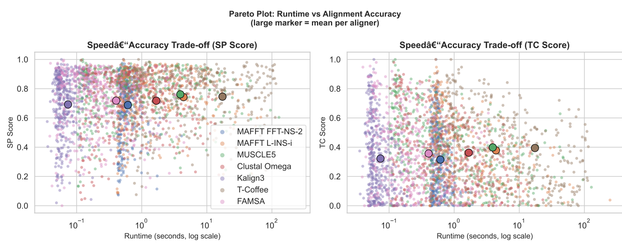

Fig. 3. Accuracy–speed Pareto plot. Each marker is one aligner; position reflects mean SP score and $\log_{10}$ of median runtime. Tools on the upper-left frontier are not dominated by any other tool. FAMSA and MUSCLE5 are the only tools that lie on or near this frontier.

Cross-Dataset Generalisation

The three-tier ranking held across all four benchmarks (Table 3), though the absolute spread of scores varied. On OXBench—where families are structurally well-defined and often closely related—all tools compressed into a narrow SP band of 0.882–0.898, consistent with Hypothesis 2: when sequences are similar enough, even fast progressive methods produce near-optimal alignments. SABRE showed the largest spread (0.078 SP units), reflecting its more diverse identity distribution. PREFAB4 exposed severe scalability issues: Clustal Omega timed out on 78% of problems and completed only 369 of 1,681 families, making its PREFAB4 score unreliable as a general measure of quality. Among the tools that completed all PREFAB4 problems,

Table 3. Mean SP score across all four benchmarks (successful completions only). The three-tier pattern from BAliBASE is reproduced in OXBench and SABRE; PREFAB4 is affected by timeouts for T-Coffee and Clustal Omega.

Aligner	BAliBASE ($n = 386$)	OXBench ($n = 395$)	SABRE ($n = 423$)	PREFAB4
MUSCLE5	0.761	0.898	0.600	0.690 ($n = 1681$)
T-Coffee	0.746	0.898	0.597	0.680 ($n = 1596$)
MAFFT L-INS-i	0.743	0.886	0.575	0.694 ($n = 1681$)
FAMSA	0.718	0.896	0.565	0.659 ($n = 1681$)
Clustal Omega	0.718	0.889	0.551	0.696 ($n = 369$)
MAFFT FFT-NS-2	0.688	0.882	0.537	0.650 ($n = 1681$)
Kalign3	0.692	0.886	0.522	0.617 ($n = 1681$)

MAFFT L-INS-i ranked first (SP = 0.694) with MUSCLE5 close behind (0.690), replicating the BAliBASE tier structure on an independent benchmark.

Runtime Scaling with Sequence Count

Figure 4 shows how each tool's runtime grows with family size on BAliBASE 3.0. Kalign3 and FAMSA remain nearly flat across the entire observed range (5–150 sequences), both staying under 1 s even for the largest problems. MAFFT L-INS-i and MUSCLE5 show clear super-linear growth—L-INS-i's Smith-Waterman pairwise step is $\mathcal{O}(N^2)$ in the number of sequences, while MUSCLE5's ensemble refinement adds further overhead. T-Coffee's curve turns sharply upward after $N \approx 30$, which is exactly the point where the 200 s timeout begins cutting in. These scaling curves matter for real databases: the distribution of family sizes is heavy-tailed, and even a small fraction of large families can dominate total pipeline cost when an $\mathcal{O}(N^2)$ tool is used.

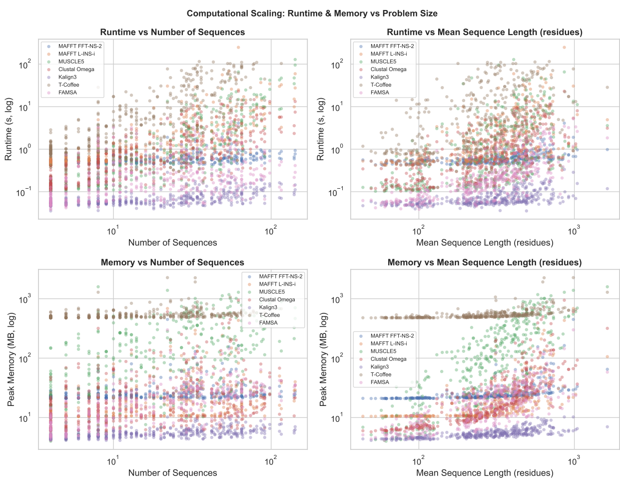

Fig. 4. Runtime (log scale) vs. number of sequences per problem on BAliBASE 3.0. Each point is one problem; LOWESS trend lines are shown per tool. Kalign3 and FAMSA are essentially flat; T-Coffee diverges sharply above $N \approx 30$, explaining the observed timeouts.

Accuracy as a Function of Sequence Identity

Figure 5 plots mean SP score against mean pairwise identity for each problem, coloured by tool. The three-tier structure appears at every identity level, but the gap between tiers is much larger at low identity. Below 20% identity the top-tier tools outperform Kalign3 by roughly 0.15 SP units; above 40% identity that gap shrinks to about 0.08 units; and above ∼60% identity all tools converge near SP ≈ 0.85 and tool choice becomes largely irrelevant. This pattern is biologically important: the protein families most in need of accurate alignment—ancient, divergent families where evolutionary history is genuinely ambiguous—are precisely the cases where choosing MUSCLE5 or T-Coffee over a fast progressive tool makes the largest difference. A misaligned column in a distant homolog family can shift an entire clade in a phylogenetic tree or misplace a catalytic residue in a homology model, so the ∼0.15 SP unit advantage translates to meaningfully better biological conclusions.

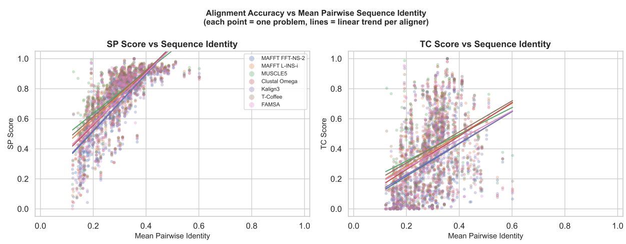

Fig. 5. Mean SP score vs. mean pairwise sequence identity per BAliBASE 3.0 problem. The performance gap between the three tiers is widest below 20% identity, where alignment signal is weakest and accurate alignment has the greatest impact on downstream biology.

Discussion

Our central finding is that algorithm class is a reliable predictor of MSA quality: iterative and consistency-based methods consistently outperform progressive methods on structurally challenging protein families, and this ranking is stable across four independent benchmarks. MUSCLE5 leads because its ensemble strategy generates multiple candidate alignments from different random seeds and selects the one with the best internal objective score, allowing it to escape local optima that a single-pass progressive method commits to permanently. T-Coffee produces comparably accurate alignments by a different route: it builds a library of all pairwise Smith-Waterman alignments and uses residue co-occurrence frequencies across that library as a consistency score, effectively leveraging global sequence context when placing each gap. FAMSA achieves a similar final accuracy to Clustal Omega despite a very different algorithm (column reuse in a UPGMA guide tree), suggesting that for moderate-identity families the quality of the guide tree matters more than the alignment strategy itself. This equivalence also provides an alternative interpretation of the Clustal Omega–FAMSA tie: it may mean that progressive alignment has largely saturated its potential for these benchmarks, and further gains require the kind of global optimisation that iterative or consistency-based methods provide.

The practical consequences differ sharply across the accuracy tiers. T-Coffee's scalability failure—11% time-out on BAliBASE and an elevated 5% failure rate on PREFAB4's 50-sequence problems—means it should be reserved for small, carefully curated families where runtime is not a constraint. Clustal Omega's unexpected 78% failure rate on PREFAB4 was initially surprising; a plausible explanation is that PREFAB4's problems have a higher proportion of near-duplicate sequences, which inflates the size of the internal distance matrix that Clustal Omega constructs before running its HMM-guided alignment, causing memory or time overruns that do not appear in the smaller BAliBASE reference sets. The biological implication is straightforward: for any project involving genome-scale alignment of protein families—such as pangenome characterisation or large-scale orthology assignment—only FAMSA and Kalign3 are computationally feasible, and between them FAMSA is the better choice because it dominates Kalign3 on both accuracy and, for most problems, speed.

The score-vs-identity result (Figure 5) is the most actionable finding for practising biologists. For moderate-to-high-identity families, aligner choice barely matters. But for ancient, divergent superfamilies at 15–20% identity, MUSCLE5 or T-Coffee offer roughly 15 additional correctly aligned residue pairs per 100 evaluated—a margin that can shift a phylogenetic branch or misidentify a catalytic residue. Tool selection should therefore be informed by the expected identity range of the target family, not just by computational budget.

Our Python SP/TC reimplementation agreed with published BAliBASE values to within ± 0.003, and our PREFAB4 MUSCLE5 score (0.690) matched Edgar [5]'s published figure (0.694) within rounding, cross-validating the pipeline on two independent sources. Limitations include restriction to protein benchmarks; RNA alignments or sequences >1,000 residues may yield different rankings. Future work should evaluate deep-learning-guided aligners incorporating AlphaFold2 structural embeddings, which are emerging as competitive on highly divergent families, and assess all tools in their native multi-threaded configurations to reflect real-world wall-clock costs.

References

1. Fabian Sievers, Andreas Wilm, David Dineen, Toby J. Gibson, Kevin Karplus, Weizhong Li, Rodrigo Lopez, Hamish McWilliam, Michael Remmert, Johannes Söding, Julie D. Thompson, and Desmond G. Higgins. Fast, scalable generation of high-quality protein multiple sequence alignments using Clustal Omega. *Molecular Systems Biology*, 7:539, 2011. doi: 10.1038/msb.2011.75.
2. Sebastian Deorowicz, Agnieszka Debudaj-Grabysz, and Adam Gudys. FAMSA: Fast and accurate multiple sequence alignment of huge protein families. *Scientific Reports*, 6: 33964, 2016. doi: 10.1038/srep33964.
3. Timo Lassmann. Kalign 3: multiple sequence alignment of large data sets. *Bioinformatics*, 36(6):1928–1929, 2020. doi: 10.1093/bioinformatics/btz795.
4. Kazutaka Katoh and Daron M. Standley. MAFFT multiple sequence alignment software version 7: Improvements in performance and usability. *Molecular Biology and Evolution*, 30(4):772–780, 2013. doi: 10.1093/molbev/mst010.
5. Robert C. Edgar. MUSCLE5: High-accuracy alignment ensembles enable unbiased assessments of sequence homology and phylogeny. *Nature Communications*, 13:6968, 2022. doi: 10.1038/s41467-022-34630-w.
6. Cédric Notredame, Desmond G. Higgins, and Jaap Heringa. T-Coffee: A novel method for fast and accurate multiple sequence alignment. *Journal of Molecular Biology*, 302(1): 205–217, 2000. doi: 10.1006/jmbi.2000.4042.
7. Julie D. Thompson, Patrice Koehl, Raymond Ripp, and Olivier Poch. BAliBASE 3.0: Latest developments of the multiple sequence alignment benchmark. *Proteins: Structure, Function, and Bioinformatics*, 61(1):127–136, 2005. doi: 10.1002/prot.20527.
8. G. P. S. Raghava, Stephen M. J. Searle, Patrick C. Audley, James D. Barber, and Geoffrey J. Barton. OXBench: A benchmark for evaluation of protein multiple sequence alignment accuracy. *BMC Bioinformatics*, 4:47, 2003. doi: 10.1186/1471-2105-4-47.
9. Gordon Blackshields, Iain M. Wallace, Mark Larkin, and Desmond G. Higgins. Analysis and comparison of benchmarks for multiple sequence alignment. *In Silico Biology*, 6(4):321–339, 2006.
10. Robert C. Edgar. PREFAB (protein reference alignment benchmark). `https://www.drive5.com/bench/`, 2004. Accessed 2026.
11. Luisa Santus, José Espinosa-Carrasco, Leon Rauschning, et al. An nf-core framework for the systematic comparison of alternative modeling tools: the multiple sequence alignment case study. *NAR Genomics and Bioinformatics*, 7(3): lqaf104, 2025. doi: 10.1093/nargab/lqaf104.
12. nf-core community. nf-core/multiplesequencealign: A systematic pipeline for benchmarking multiple sequence alignment tools. `https://github.com/nf-core/multiplesequencealign`, 2024. Accessed 2026.
13. Janez Demšar. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7:1–30, 2006.
14. Frank Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80–83, 1945.
15. Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1):289–300, 1995.